

DSA 8020 Project 1

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```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)  
library(MASS)
```

```
##  
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':  
##  
##   select
```

```
library(mgcv)
```

```
## Loading required package: nlme
```

```
##  
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':  
##  
##   collapse
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
library(car)
```

```
## Loading required package: carData
```

```
##  
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':  
##  
##   recode
```

Instructions

For each problem, your answer should include the following elements:

- A descriptive summary of the data using summary statistics and at least one plot/graph or table. Give your description in paragraph form.

- An appropriate inferential analysis to address the question. This may be a confidence interval, hypothesis test, or both. You can use the α value/confidence level of your choice. If you perform a hypothesis test, state the hypotheses, test statistic, p-value, decision, and conclusion. If you create a confidence interval, state the confidence level and provide an interpretation of the interval.
 - A thoughtful conclusion that summarizes your findings. You should also discuss any limitations or complications of your analysis. This should include the following:
 - An assessment of whether the modeling assumptions are reasonable.
 - Discussion of any unusual features of the data. For example, do outliers/high leverage points affect your results?
-

Problem 1: Gala

- Perform a detailed regression analysis using the gala data set. Use Endemics as the response variable, excluding Species from this analysis, to address the following questions:

Code:

```
library(faraway)

##
## Attaching package: 'faraway'

## The following objects are masked from 'package:car':
##
##   logit, vif

data(gala)
```

Data Format

Source: M. P. Johnson and P. H. Raven (1973) “Species number and endemism: The Galapagos Archipelago revisited” Science, 179, 893-895

The dataset contains the following variables

Species - the number of plant species found on the island

Endemics - the number of endemic species

Area - the area of the island (km²)

Elevation - the highest elevation of the island (m)

Nearest - the distance from the nearest island (km)

Scruz - the distance from Santa Cruz island (km)

Adjacent - the area of the adjacent island (square km)

1. Provide a descriptive summary of the data set.

Code:

```
summary(gala)
```

```
##      Species      Endemics      Area      Elevation
## Min.   : 2.00   Min.   : 0.00   Min.   : 0.010   Min.   : 25.00
## 1st Qu.: 13.00  1st Qu.: 7.25   1st Qu.: 0.258   1st Qu.: 97.75
## Median : 42.00  Median :18.00   Median : 2.590   Median : 192.00
## Mean   : 85.23  Mean   :26.10   Mean   : 261.709   Mean   : 368.03
## 3rd Qu.: 96.00  3rd Qu.:32.25   3rd Qu.: 59.237   3rd Qu.: 435.25
## Max.   :444.00  Max.   :95.00   Max.   :4669.320   Max.   :1707.00
##      Nearest      Scrutz      Adjacent
## Min.   : 0.20   Min.   : 0.00   Min.   : 0.03
## 1st Qu.: 0.80   1st Qu.: 11.03  1st Qu.: 0.52
## Median : 3.05   Median : 46.65  Median : 2.59
## Mean   :10.06   Mean   : 56.98  Mean   : 261.10
## 3rd Qu.:10.03   3rd Qu.: 81.08  3rd Qu.: 59.24
## Max.   :47.40   Max.   :290.20  Max.   :4669.32
```

```
head(gala)
```

```
##      Species Endemics Area Elevation Nearest Scrutz Adjacent
## Baltra      58      23 25.09      346      0.6  0.6      1.84
## Bartolome   31      21 1.24      109      0.6 26.3     572.33
## Caldwell     3       3 0.21      114      2.8 58.7      0.78
## Champion    25       9 0.10       46      1.9 47.4      0.18
## Coamano      2       1 0.05       77      1.9 1.9     903.82
## Daphne.Major 18      11 0.34      119      8.0 8.0      1.84
```

2. Is at least one of the predictors useful in predicting the response?

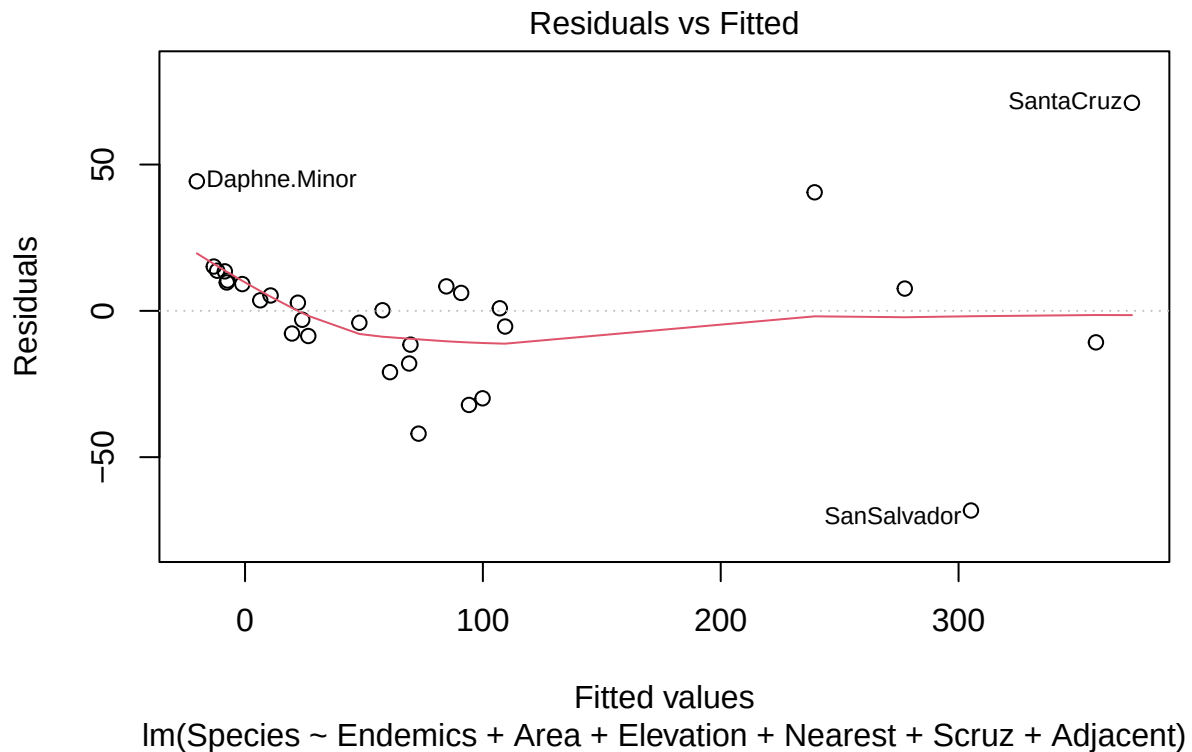
Code:

```
# getting p-vals
model <- lm(Species ~ Endemics + Area + Elevation + Nearest + Scrutz + Adjacent, data = gala)
summary(model)
```

```
##
## Call:
## lm(formula = Species ~ Endemics + Area + Elevation + Nearest +
##      Scrutz + Adjacent, data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -68.219 -10.225   1.830   9.557  71.090
##
## Coefficients:
```

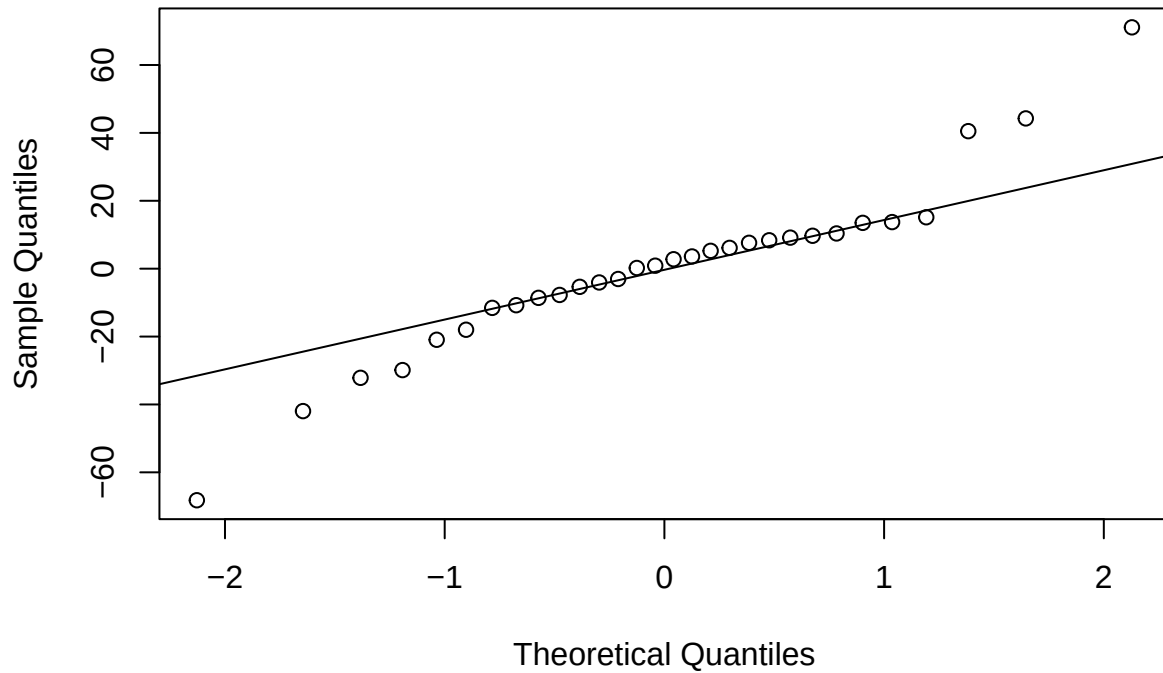
```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -15.337942   9.423550  -1.628   0.117
## Endemics     4.393654   0.481203   9.131 4.13e-09 ***
## Area         0.013258   0.011403   1.163   0.257
## Elevation    -0.047537   0.047596  -0.999   0.328
## Nearest     -0.101460   0.500871  -0.203   0.841
## Scruz        0.008256   0.105884   0.078   0.939
## Adjacent     0.001811   0.011879   0.152   0.880
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 28.96 on 23 degrees of freedom
## Multiple R-squared:  0.9494, Adjusted R-squared:  0.9362
## F-statistic: 71.88 on 6 and 23 DF,  p-value: 9.674e-14
```

```
# residual vs fitted
plot(model, which = 1)
```



```
# qq plot
qqnorm(resid(model))
qqline(residuals(model))
```

Normal Q-Q Plot



```
# correlation
cor(gala[, c("Endemics", "Area", "Elevation", "Nearest", "Scruz", "Adjacent")], gala["Species"])
```

```
##           Species
## Endemics  0.97087652
## Area      0.61784307
## Elevation 0.73848666
## Nearest   -0.01409407
## Scruz     -0.17114244
## Adjacent  0.02616635
```

Answer:

- Allowing an alpha of 0.05, we can see that Endemics is useful in predicting the response as it's p-value is 4.13e-09.
- Residual plot shows no significant patterns or heteroscedasticity; doesn't violate assumptions of linear regression.

3. Perform model selection to determine whether all the predictors help to explain the response.

Code:

```
# remove pointless vars iteratively based on AIC
stepmodel <- stepAIC(model, direction = "both")
```

```
## Start: AIC=207.99
## Species ~ Endemics + Area + Elevation + Nearest + Scrutz + Adjacent
##
##           Df Sum of Sq  RSS   AIC
## - Scrutz    1         5 19300 206.00
## - Adjacent   1        19 19314 206.02
## - Nearest    1        34 19329 206.04
## - Elevation  1       837 20132 207.26
## - Area       1      1134 20429 207.71
## <none>              19295 207.99
## - Endemics   1     69937 89231 251.93
##
## Step: AIC=206
## Species ~ Endemics + Area + Elevation + Nearest + Adjacent
##
##           Df Sum of Sq  RSS   AIC
## - Adjacent   1        19 19318 204.03
## - Nearest    1        32 19332 204.05
## - Elevation  1       838 20138 205.28
## - Area       1      1129 20429 205.71
## <none>              19300 206.00
## + Scrutz     1         5 19295 207.99
## - Endemics   1     74567 93867 251.45
##
## Step: AIC=204.03
## Species ~ Endemics + Area + Elevation + Nearest
##
##           Df Sum of Sq  RSS   AIC
## - Nearest    1         50 19369 202.11
## <none>              19318 204.03
## - Area       1      1376 20694 204.09
## - Elevation  1      2232 21550 205.31
## + Adjacent   1         19 19300 206.00
## + Scrutz     1          4 19314 206.02
## - Endemics   1    150599 169918 267.26
##
## Step: AIC=202.11
## Species ~ Endemics + Area + Elevation
##
##           Df Sum of Sq  RSS   AIC
## <none>              19369 202.11
## - Area       1      1498 20866 202.34
## - Elevation  1      2287 21655 203.45
## + Nearest    1         50 19318 204.03
## + Adjacent   1         37 19332 204.05
## + Scrutz     1          8 19360 204.09
## - Endemics   1    150578 169947 265.26
```

```
summary(stepmodel)
```

```
##
## Call:
## lm(formula = Species ~ Endemics + Area + Elevation, data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -67.691 -10.530   2.387  10.529  72.723
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -15.891237   7.569210  -2.099   0.0456 *
## Endemics      4.331791   0.304686  14.217 8.97e-14 ***
## Area          0.012669   0.008936   1.418   0.1681
## Elevation    -0.041439   0.023653  -1.752   0.0916 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.29 on 26 degrees of freedom
## Multiple R-squared:  0.9492, Adjusted R-squared:  0.9433
## F-statistic: 161.8 on 3 and 26 DF,  p-value: < 2.2e-16
```

Answer:

- Using stepwise AIC selection, we can get a final optimal model of

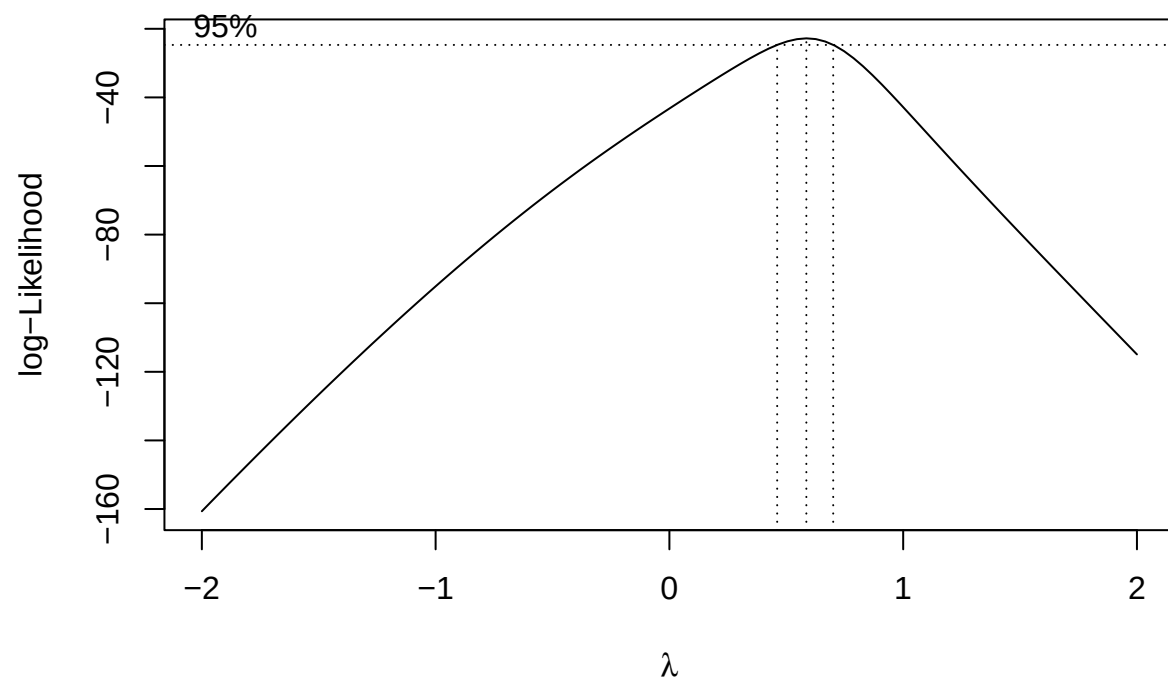
Species ~ Endemics + Area + Elevation

- Endemics: Significant positive effect on species count (p-val=8.97e-14, t-val=14.217).
- Area: Weak but visible positive association with species diversity (p-value=0.168). Shows larger areas
- Elevation: Very slight negative relationship with species diversity (p-val=0.092).

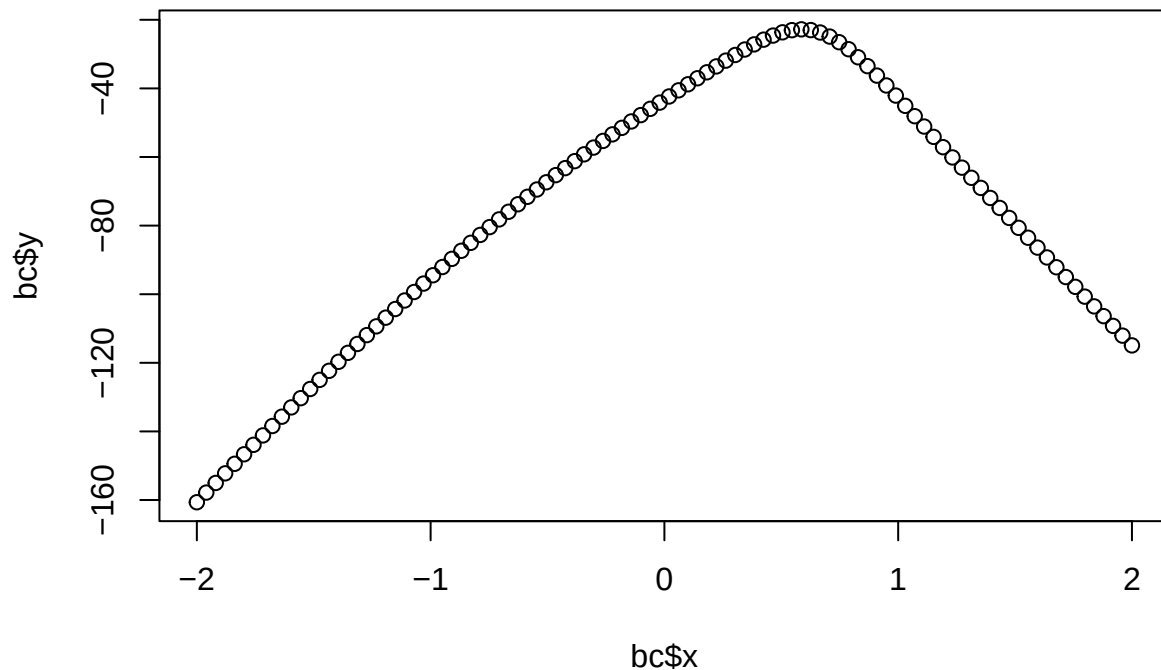
4. Perform a Box-Cox transformation to determine if there is a need to transform the response.

Code:

```
bc <- boxcox(stepmodel)
```

```
plot(bc)
```



- As we get an approximate λ value of 0.5, we can try a square root transformation.

```
bc_sqrt_species <- sqrt(gala$Species)
bc_sqrt_model <- lm(bc_sqrt_species ~ Endemics + Area + Elevation, data = gala)
summary(bc_sqrt_model)
```

```
##
## Call:
## lm(formula = bc_sqrt_species ~ Endemics + Area + Elevation, data = gala)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-3.02445	-0.72222	-0.03576	0.54130	2.48345

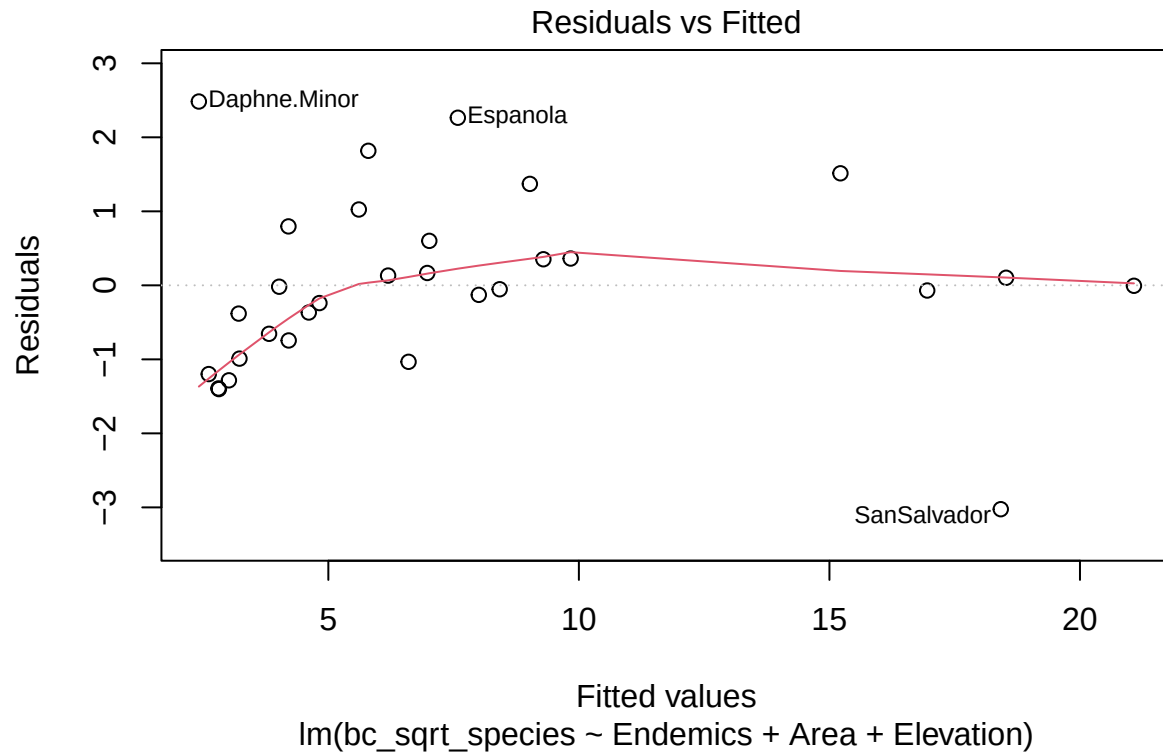
```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.4059105	0.3467849	6.938	2.29e-07 ***
Endemics	0.1992437	0.0139592	14.273	8.19e-14 ***
Area	-0.0003834	0.0004094	-0.937	0.358
Elevation	0.0001038	0.0010837	0.096	0.924

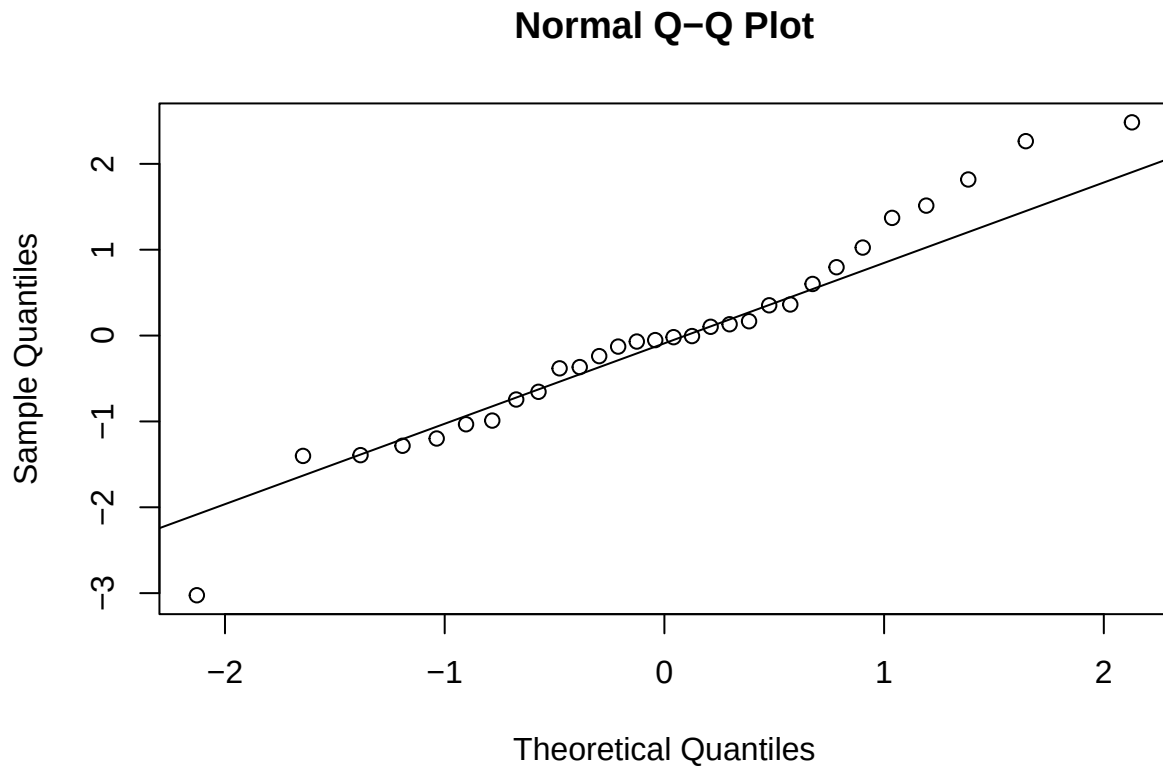
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.25 on 26 degrees of freedom
## Multiple R-squared:  0.9521, Adjusted R-squared:  0.9466
```

```
## F-statistic: 172.4 on 3 and 26 DF,  p-value: < 2.2e-16
```

```
# residual vs fitted  
plot(bc_sqrt_model, which = 1)
```



```
# normal QQ  
qqnorm(resid(bc_sqrt_model))  
qqline(residuals(bc_sqrt_model))
```



- Both plots show significant improvements over our original model. There is significantly less spread in Residuals as well as a more linear relationship in our QQ plot.
- Residual plots for both models(original and transformed) showed no significant patterns or heteroscedasticity; transformation doesn't violate assumptions of linear regression.
- R-squared increased slightly to 0.9521.
- Marginal improvement in adjusted R-squared to 0.9466.

5. Compare a generalized additive model fit with the linear regression fit and explain why you preferred one model over the other.

Code:

```
gam_model <- gam(bc_sqrt_species ~ s(Endemics) + s(Area) + s(Elevation), data = gala)
AIC(bc_sqrt_model, gam_model)
```

```
##           df      AIC
## bc_sqrt_model 5.000000 104.25466
## gam_model     6.086706  99.82638
```

```
BIC(bc_sqrt_model, gam_model)
```

```
##                df      BIC
## bc_sqrt_model  5.000000 111.2606
## gam_model      6.086706 108.3551
```

Answer:

- Our degrees of freedom is higher for the generalized additive fit model (6.08), proving it to be the better fit.
 - GAM model also shows to perform better with both Akaike (99.8) and Bayesian (108.3) criterion values being lower than observed with the Box-Cox fit alone (A: 104.2, B: 111.2).
 - Thus, our GAM model is preferred.
-

6. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.

Code:

```
# comparing with Box-Cox sqrt model
lm_dev <- deviance(bc_sqrt_model)
null_lm_dev <- deviance(update(bc_sqrt_model, ~ -1))
dev_explained_lm <- 1 - (lm_dev / null_lm_dev)
dev_explained_lm
```

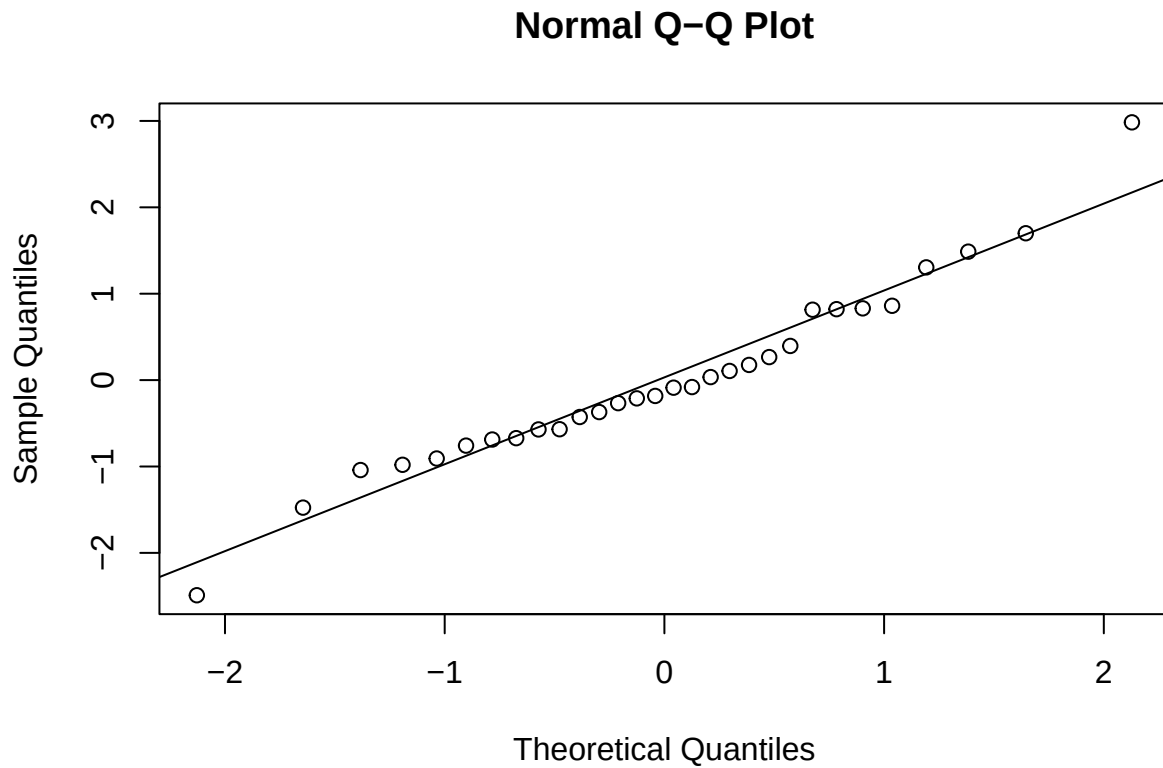
```
## [1] 0.9841002
```

```
# gam model
gam_dev <- deviance(gam_model)
null_gam_model <- gam(bc_sqrt_species ~ 1, data = gala)
null_gam_dev <- deviance(null_gam_model)
dev_explained_gam <- 1 - (gam_dev / null_gam_dev)
dev_explained_gam
```

```
## [1] 0.9616005
```

- Although there is a slight drop in deviance explained, (0.96 as opposed to 0.98 with Box-Cox sqrt), I still prefer the GAM model due to its improvement in both AIC and BIC as well as its improved flexibility shown by its degree of freedom (6).

```
# normal QQ
qqnorm(residuals(gam_model))
qqline(residuals(gam_model))
```



- Our QQ plots show our residuals to be normally distributed, showing no violation in model assumptions.

7. Predict the response value when all the chosen predictors take their median value. Quantify the prediction uncertainty.

Code:

8. Provide a thoughtful conclusion that summarize your findings.

Code:

Problem 2: Salaries

- Perform a detailed analysis using the Salaries data set to comment on the salary differences between male and female faculty members.

Code:

```
library(carData)
data(Salaries)
```

Data Format

References: Fox J. and Weisberg, S. (2019) An R Companion to Applied Regression, Third Edition, Sage.

A data frame with 397 observations on the following 6 variables.

rank - a factor with levels AssocProf AsstProf Prof

discipline - a factor with levels A (“theoretical” departments) or B (“applied” departments).

yrs.since.phd - years since PhD.

yrs.service - years of service.

sex - a factor with levels Female Male

salary - nine-month salary, in dollars.

1. Provide a descriptive summary of the data set.

Code:

```
head(Salaries)
```

```
##      rank discipline yrs.since.phd yrs.service sex salary
## 1    Prof         B           19         18 Male 139750
## 2    Prof         B           20         16 Male 173200
## 3  AsstProf         B            4           3 Male  79750
## 4    Prof         B           45         39 Male 115000
## 5    Prof         B           40         41 Male 141500
## 6 AssocProf         B            6           6 Male  97000
```

```
summary(Salaries)
```

```
##      rank      discipline yrs.since.phd   yrs.service      sex
## AsstProf : 67  A:181      Min.   : 1.00   Min.   : 0.00  Female: 39
## AssocProf: 64  B:216      1st Qu.:12.00   1st Qu.: 7.00   Male  :358
## Prof      :266                Median :21.00   Median :16.00
##                Mean   :22.31   Mean   :17.61
##                3rd Qu.:32.00   3rd Qu.:27.00
##                Max.   :56.00   Max.   :60.00
##      salary
## Min.   : 57800
## 1st Qu.: 91000
## Median :107300
## Mean   :113706
## 3rd Qu.:134185
## Max.   :231545
```

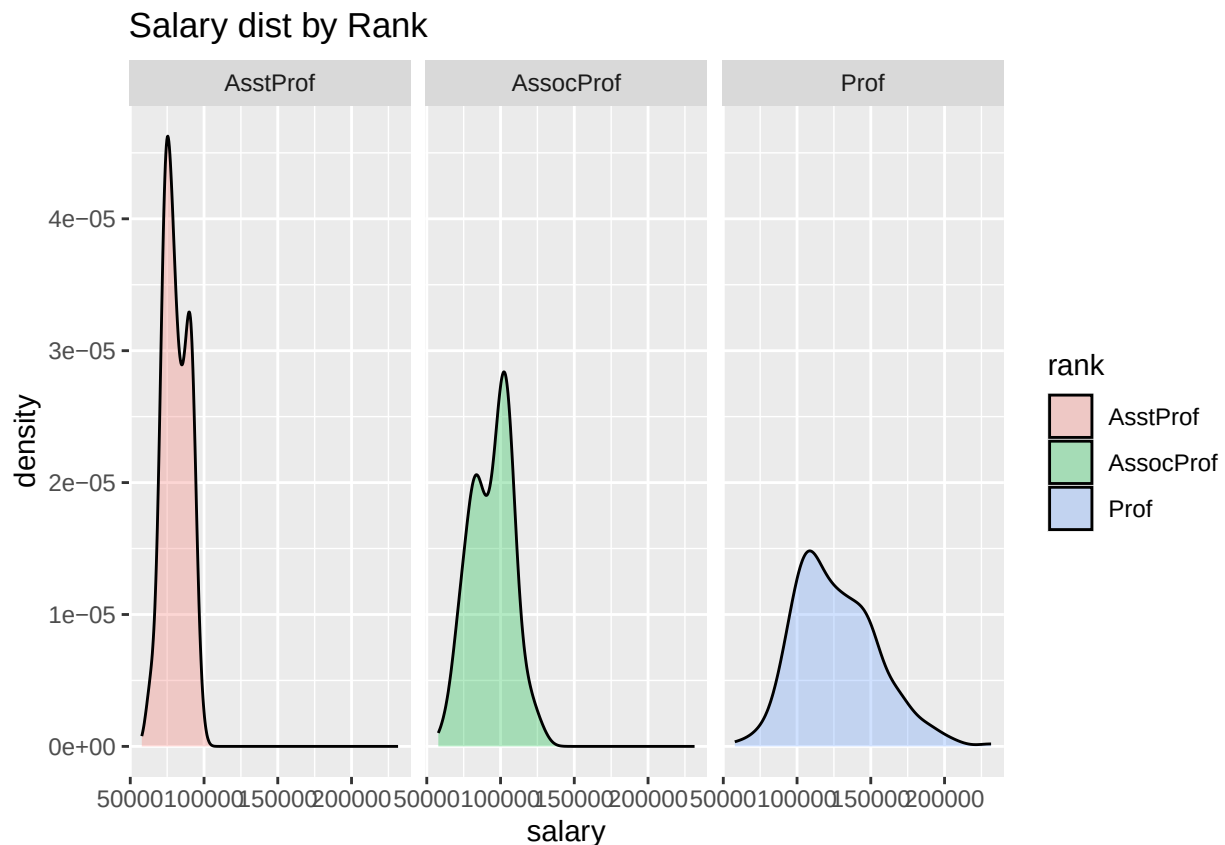
Answer:

- The majority of the staff roster is made up of Full Professors (266).
 - There are slightly more Applied discipline (216) than Theoretical (181).
 - Mean years since Phd is 22.31 years, while mean years of service is 17.61 years.
 - Male staff makes up an overwhelming majority of the staff roster.
 - Salary varies widely, with a median of 107k, and a maximum of 231k.
 - Mean salary is slightly higher at 113k -> potential positive skewing salary distribution.
 - Distribution indicates some professors earn significantly more, most likely based on rank and years of service.
-

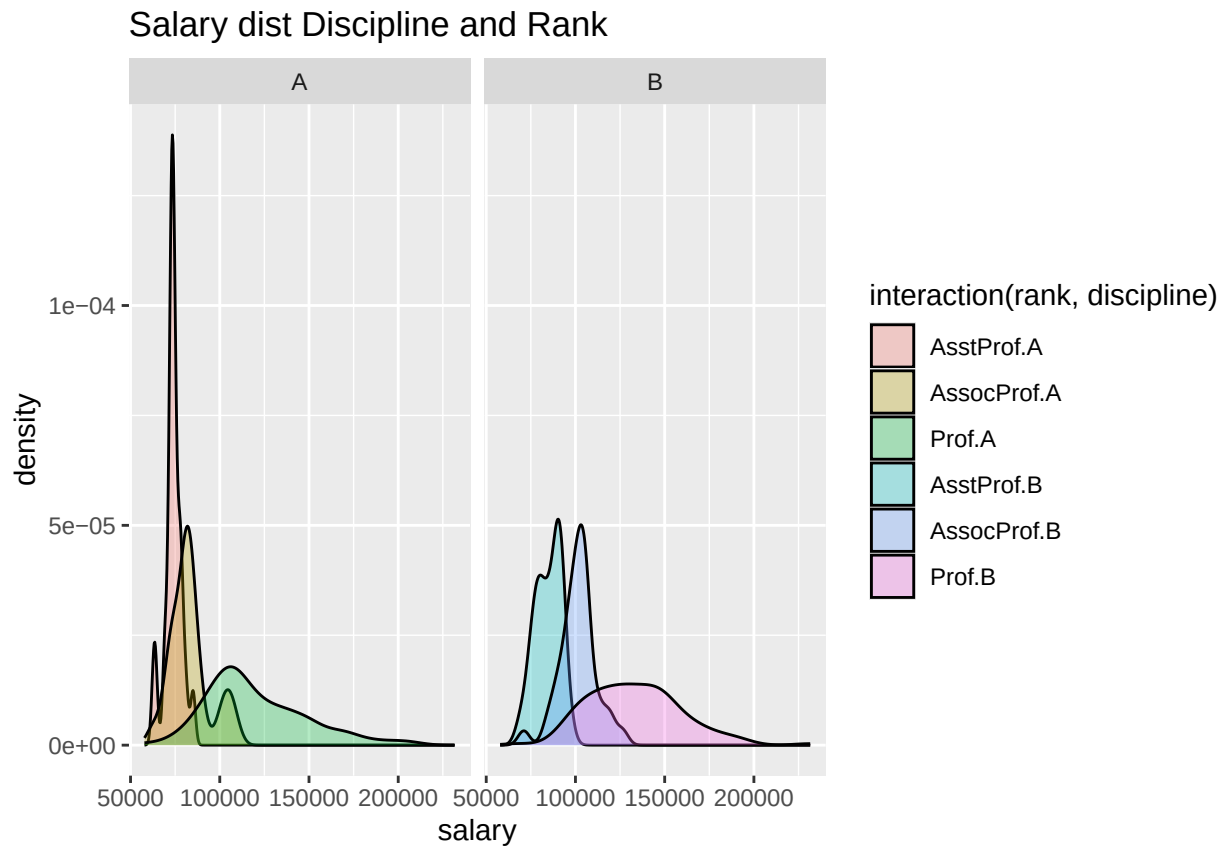
2. Comment on whether there is a need to include interactions between these categorical predictors. Select your model and justify your choice.

Code:

```
ggplot(Salaries, aes(x = salary, fill = rank)) +  
  geom_density(alpha = 0.3) +  
  facet_wrap(~rank) +  
  labs(title = "Salary dist by Rank", x = "salary", y = "density")
```

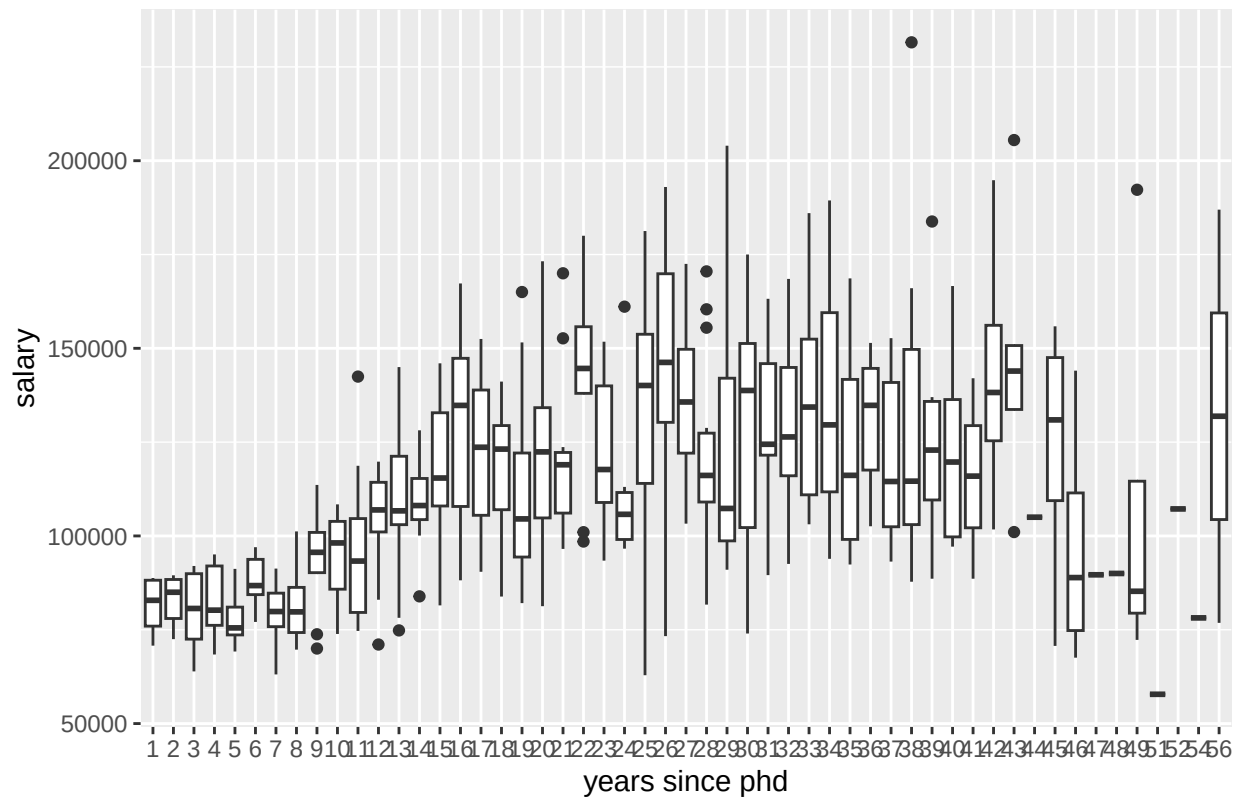



```
# salary distribution by discipline and rank
ggplot(Salaries, aes(x = salary, fill = interaction(rank, discipline))) +
  geom_density(alpha = 0.3) +
  facet_wrap(~discipline) +
  labs(title = "Salary dist Discipline and Rank", x = "salary", y = "density")
```

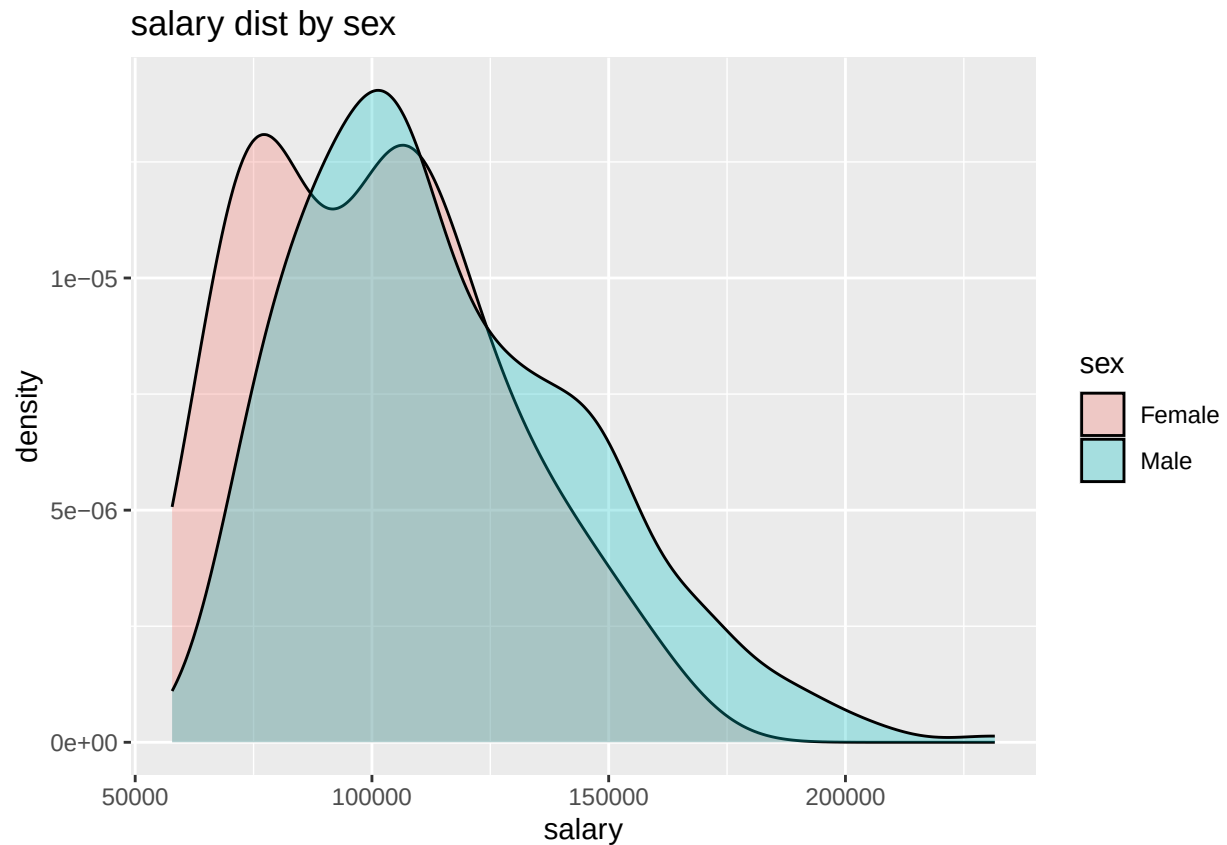


```
# salary distribution years since pdf (cat)
ggplot(Salaries, aes(x = factor(yrs.since.phd), y = salary)) +
  geom_boxplot() +
  labs(title = "Salary dist by yrs.since.phd", x = "years since phd", y = "salary")
```

Salary dist by yrs.since.phd



```
# salary dist by sex
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  labs(title = "salary dist by sex", x = "salary", y = "density")
```



3. Perform a log transformation on response to determine if there is any improvement in terms of model fit.

Code:

4. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.

Code:

5. Write a thoughtful conclusion to comment on the salary differences between male and female faculty.

Code: